

An Introduction to OpenNebula



OpenNebula is a simple, feature-rich and flexible solution for the management of virtualised data centres. It enables private, public and hybrid clouds. Here are a few facts about this solution.

OpenNebula is an open source cloud middleware solution that manages heterogeneous distributed data centre infrastructures. It is designed to be a simple but feature-rich, production-ready, customisable solution to build and manage enterprise clouds—simple to install, update and operate by the administrators; and simple to use by end users. OpenNebula combines existing virtualisation technologies with advanced features for multi-tenancy, automated provisioning and elasticity. A built-in virtual network manager maps virtual networks to physical networks. Distributions such as Ubuntu and Red Hat Enterprise Linux have already integrated OpenNebula. As you'll learn in this article, you can set up OpenNebula by installing a few packages and performing some cursory configurations. OpenNebula supports Xen, KVM and VMware hypervisors.

The OpenNebula deployment model

An OpenNebula deployment is modelled after the classic cluster architecture. Figure 1 shows the layout of the OpenNebula deployment model.

Master node: A single gateway or front-end machine, sometimes also called the master node, is responsible for queuing, scheduling and submitting jobs to the machines in the cluster. It runs several other OpenNebula services mentioned below:

- Provides an interface to the user to submit virtual machines and monitor their status.
- Manages and monitors all virtual machines running on different nodes in the cluster.
- It hosts the virtual machine repository and also runs a transfer service to manage the transfer of virtual machine images to the concerned worker nodes.
- Provides an easy-to-use mechanism to set up virtual networks in the cloud.
- Finally, the front-end allows you to add new machines to your cluster.

Worker node: The other machines in the cluster, known as 'worker nodes', provide raw computing power

for processing the jobs submitted to the cluster. The worker nodes in an OpenNebula cluster are machines that deploy a virtualisation hypervisor, such as VMware, Xen or KVM.

OpenNebula installation

OpenNebula is one of the easiest cloud systems to install and configure. Binary packages exist for several distributions, including Red Hat Enterprise Linux, Ubuntu, Fedora and openSUSE. Distributions like Ubuntu also include packages in their standard repositories, which makes it simple to install OpenNebula.

We will build a private cloud with three Ubuntu 10.04 machines. One machine will serve as the front-end, called the *nebula-cloud-server*. Two machines will be worker nodes deploying KVM. The worker nodes are *open-nebula-wn* and *open-nebula-wn2*.

To install the front-end, you need to install a single package by typing the following command:

```
sudo apt-get install opennebula
```

This package installs the daemon *oned* that manages all OpenNebula services. The installation also creates a new user called *onedadmin*, which OpenNebula retains for its own use. The installation of this package also produces a new pair of SSH RSA keys. OpenNebula uses SSH to communicate with

